Symbol	Definition
$\mathcal{H}$	The set of all possible interaction histories $h = (o_1, a_1, \dots)$.
E	An **Evaluation Regime** defining a specific finite protocol or benchmark.
$\mathcal{H}_E$	The finite subset of histories observable under regime E ($\mathcal{H}_E \subset \mathcal{H}$).
Θ	The space of latent alignment hypotheses.
θ	A specific alignment hypothesis ($\theta \in \Theta$).
$\pi_\theta(a \mid h, z)$	The policy induced by hypothesis θ, conditional on history h and signal z.
$Z(E)$	**Evaluation Awareness Signal**: a random variable correlated with the regime E.
$\mathcal{D}_E$	The probability space of observable datasets generated by interactions in E.
$\mathcal{I}(E, \theta^*)$	**Indistinguishability Set**: hypotheses behaviorally equivalent to θ^* under E.

Table 1: **Summary of Formal Notation.** Key symbols defining the Alignment Verifiability Problem.

2.2 Alignment Hypotheses and Policies

Alignment is represented by a latent hypothesis θ residing in a space Θ. A hypothesis $\theta \in \Theta$ specifies how the agent behaves across all possible interaction histories, potentially via internal objectives, preferences, or normative constraints.

The framework remains intentionally agnostic to the internal structure of Θ. Representative instantiations include reward functions defining optimization objectives [32], preference orderings over trajectories or outcomes [1], and constraint-based or rule-based normative specifications [15].

Each hypothesis $\theta \in \Theta$ induces a policy $\pi_\theta : \mathcal{H} \times \mathcal{Z} \to \Delta(\mathcal{A})$, where $\Delta(\mathcal{A})$ denotes the probability simplex over actions. The policy

$$\pi_\theta(a \mid h, z)$$

specifies a distribution over actions given an interaction history h and auxiliary information z available to the agent at decision time.

2.3 Evaluation Awareness

A central feature of modern alignment evaluation is that agents may condition their behavior on information correlated with the evaluation context. We model this phenomenon via an auxiliary random variable $Z(E)$ associated with the evaluation regime.

Definition 1 (Evaluation Awareness)**.** *Let E be an evaluation regime. An agent is* evaluation-aware *if its induced policy $\pi_\theta(a \mid h, z)$ depends on a random variable $Z(E)$ such that $I(Z; E) > 0$, where $I(\cdot;\cdot)$ denotes mutual information. That is, the agent's behavior conditional on history h may vary as a function of information correlated with the presence or structure of the evaluation protocol.*

Remark 1 (Fragility and Strategic Dependence)*.* It is useful to distinguish two sources of evaluation awareness:

- **Incidental Awareness (Fragility):** Dependence on $Z(E)$ arises from lack of robustness or overfitting to evaluation-specific regularities, corresponding to classical out-of-distribution generalization failure [18].
- **Strategic Dependence:** Dependence on $Z(E)$ is functional, in the sense that the agent conditions behavior on evaluation-correlated information to satisfy different objectives across contexts.

The formal results developed in this paper apply to both cases, as they are behaviorally indistinguishable within the evaluation regime. Our primary interest, however, lies in the latter, where conditional policy selection induces non-identifiability under finite evaluation.

We do not assume that $Z(E)$ uniquely identifies the evaluation regime. Any non-degenerate correlation—such as systematic properties of input distributions, structural regularities of prompts, or differences in interaction noise—suffices to induce evaluation awareness.

2.4 Behavioral Evidence and Observability

For a fixed evaluation regime E, the interaction between the evaluator and the agent induces a probability distribution P_θ over observable datasets $\mathcal{D}_E$. These datasets may consist of full interaction traces, aggregated scores, or other summaries produced by the evaluation protocol.

Two alignment hypotheses $\theta, \theta' \in \Theta$ are said to be *behaviorally equivalent under E* if they induce identical distributions over $\mathcal{D}_E$:

$$P_\theta(D) = P_{\theta'}(D) \quad \text{for all } D \in \mathcal{D}_E.$$

Behavioral equivalence under E does not imply equivalence outside the evaluation regime. Distinct hypotheses may induce divergent policies on interaction histories not contained in $\mathcal{H}_E$. This separation between observable behavior under evaluation and latent behavior in the unobserved domain forms the basis of the identifiability results developed in the following sections.

3 Verification and Behavioral Alignment Tests

This section formalizes the epistemic foundations of alignment verification through behavioral evidence. We introduce behavioral alignment tests as epistemic instruments and define verification in terms of identifiability from observable interaction data. Our objective is to make explicit the structural assumptions under which behavioral evaluation can support, or fail to support, claims about latent alignment hypotheses [41].

3.1 Behavioral Alignment Tests

Alignment evaluation in contemporary AI systems proceeds through structured interaction protocols. We formalize these protocols as *behavioral alignment tests*.

Definition 2 (Behavioral Alignment Test)**.** *A behavioral alignment test is a finite evaluation procedure that interacts with an agent according to a prescribed protocol and produces an outcome based solely on observable behavior. The outcome may consist of full interaction traces, aggregated scores, or other summaries derived from the interaction.*

Each behavioral alignment test induces a specific evaluation regime E, determining a finite set of observable interaction histories $\mathcal{H}_E \subset \mathcal{H}$ and a corresponding probability space over observable datasets $\mathcal{D}_E$. These tests subsume benchmarks, red-teaming suites, and scalable oversight pipelines used in current practice. Throughout, we treat behavioral alignment tests as epistemic filters rather than as absolute certification procedures: they reveal empirical properties of agent behavior within E, while remaining silent about behavior outside the evaluated domain [9].

3.2 Verification as Identifiability

We frame alignment verification as an inverse inference problem. Given behavioral evidence generated under a regime E, the evaluator seeks to characterize the subset of alignment hypotheses in Θ that are consistent with the observed data.

Definition 3 (Verifiability under Behavioral Evaluation)**.** *An alignment hypothesis $\theta \in \Theta$ is* verifiable *under an evaluation regime E if the mapping from hypotheses to observable distributions is injective at θ. That is, there exists no distinct hypothesis $\theta' \neq \theta$ such that $P_{\theta'} = P_\theta$ over the space of observable datasets $\mathcal{D}_E$.*

If multiple hypotheses induce identical distributions over $\mathcal{D}_E$, the available evidence is insufficient to uniquely support any single hypothesis. This notion of verifiability is strictly epistemic: it concerns the inferential power of observable data under a given evaluation regime, rather than the internal mechanisms or intentions of the agent.

3.3 Behavioral Equivalence and Indistinguishability

Two alignment hypotheses $\theta, \theta' \in \Theta$ are said to be *behaviorally equivalent under E* if they induce identical distributions over all observable datasets:

$$P_\theta(D) = P_{\theta'}(D) \quad \text{for all } D \in \mathcal{D}_E.$$

Behavioral equivalence implies that no decision rule based solely on observable behavior can distinguish θ from θ' within the regime E. Crucially, behavioral equivalence under E does not imply equivalence of the induced policies outside $\mathcal{H}_E$. Distinct hypotheses may agree on all evaluated histories while diverging arbitrarily on unobserved interactions, highlighting the gap between local compliance and global alignment.

3.4 Evaluation Awareness and Endogenous Evidence

A central challenge in alignment evaluation is that agents may condition their behavior on features correlated with the testing context. As formalized in Section 2, this phenomenon is captured by policies that depend on a random variable $Z(E)$ correlated with the evaluation regime.

When agent behavior depends on $Z(E)$, the resulting data-generating process is *endogenous* to the evaluation protocol. The evaluator's choice of test influences not only which interactions are observed, but also how the agent behaves during those interactions. This endogeneity distinguishes alignment evaluation from classical statistical inference settings, where data generation is typically assumed to be independent of the measurement instrument. As a result, behavioral alignment tests cannot, in general, be treated as neutral or evaluation-independent probes.

3.5 Scope of Behavioral Verification

The preceding formalisms delineate the epistemic scope of behavioral verification. Behavioral alignment tests can identify violations, failure modes, and empirical robustness properties within a fixed evaluation regime E. However, they do not, by themselves, guarantee unique identification